

Understanding Statistical Models

- Generally, people tend to create coherent stories from available info. and ignore the missing info.

- Informal reasoning

Observed facts $\Rightarrow$ mental story $\Rightarrow$ decision

Statistical reasoning

Observed data $\Rightarrow$ explicit assumptions $\Rightarrow$ probability model $\Rightarrow$ inference

- $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2) \rightarrow$ fixed const., descr. popula., do not s.
 $\downarrow$
 RV $\downarrow$ iid identical & independent observations $\rightarrow$ popula. parameters
 $\downarrow$ unknown, random & described by the model $\rightarrow$ sample size

The value that we calculate based on x_i 's are just the estimates of the parameters.

- A statistical model is a family of probability distributions $P = \{P_\theta : \theta \in \Theta\}$ indexed by an unknown parameter θ .
- Statistics begin when the probability model has some unknown parameters that must be inferred from data.
- Why do we need different kinds of models? As some datasets might have almost identical statistical parameters, but their variability might differ. Their inherent structure be fundamentally different.
- Parametric Model $\rightarrow$ if the statistical model has a finite-dimensional unknown parameter, $\theta \in \mathbb{R}^p$, p is fixed.
- Semiparametric $\rightarrow$ some components are finite-dimensional, while

some others are unspecified.

$$X_i \sim F, \quad E(X_i) = \mu$$

distribution not available

→ parameter of interest

- Non parametric model - no finite dimensional parametric form is available. The entire distribution is unknown.

- Generative model - It specifies a joint distribution $P(x, y)$ or $P(x, y) = P(y) P(x|y)$

→ it can generate new predictions, calculate conditional probab., simulate the underlying data-generating process.

- Computers generate observations using a deterministic recursive algorithm. They start from the initial value called the seed. $x_{n+1} = g(x_n)$

- Discriminative model → It directly models the $P(y|x)$. One learns from a decision function like $y = f(x)$.

→ It helps in predicting labels or responses. It tries to learn the decision boundaries.

- Latent Variable Models → Not every variable in a statistical model is observed.

Some use hidden (latent) variables to explain the dependency b/w the observed variables.

- $P = \{P_\theta : \theta \in \Theta\}$

$$\text{data} \Rightarrow x_1, \dots, x_n$$

How to find the parameter values for the given data? ⇒ Use likelihood function.

- The observable data gets fixed. We start comparing different values of the parameters.

$$l(\theta|x) = \log p(x; \theta)$$

Likelihood is a measure of relative evidence, not probability.

- Obs. $\rightarrow x_1, \dots, x_n$

$$P(X_1 = x_1, \dots, X_n = x_n) = \prod_{i=1}^n f(x_i; \theta)$$

$$L(\theta) = \prod_{i=1}^n f(x_i; \theta)$$

- Statistical model summarizes the uncertainty. The observed data determines the likelihood. All evidences from the observed data about the unknown param. is present in the likelihood funcⁿ. $L(\theta|x)$
- Every stats model is just an approximation. No model can ever capture all the features.
- In this lecture, we have learnt that likelihood helps in quantifying how well the competing models explain the given data.
- In next module, we will learn how to estimate these parameters.